

Four questions, one reduction

Each question sets up a matrix, reduces it, and reads a different part of the result. The setup barely changes between them — what changes is where to look.

ABOUT A SINGLE VECTOR

1

1

Is **b** in the span?

consistent $\Rightarrow$ yes

$[\mathbf{v} \cdots \mathbf{v} \mid \mathbf{b}]$

Augment the spanning vectors with **b** and reduce. A row of the form $[0 \cdots 0 \mid d]$ with $d \neq 0$ is a contradiction and settles it as no; anything else means **b** is reachable, and the reduced form also hands you the coefficients that reach it.

ABOUT THE WHOLE SET

3

2

Does the set span $\mathbb{R}^n$?

a pivot in every row

$\text{rank}(A) = n$

Vectors as columns of an $n \times k$ matrix. Every row needs a pivot — a pivotless row is a direction of $\mathbb{R}^n$ nothing in the set reaches. Note this is a condition on rows, while redundancy below is a condition on columns; confusing the two is the usual error.

3

Which vectors are redundant?

a pivot in every column $\Rightarrow$ none

non-pivot columns of $\text{rref}(A)$

A column without a pivot corresponds to a vector already in the span of the earlier ones, so it can be dropped without changing the span. The reduced form names which — removing them leaves a **minimal spanning set**.

4

Is the set a basis?

a pivot in every row and column

$\det(A) \neq 0, A \text{ square}$

Both conditions at once: spanning with no redundancy. For n vectors in $\mathbb{R}^n$ the matrix is square, so the determinant test settles it in one step — which is the **invertibility equivalence** restated for spanning sets.

The common step is worth stating plainly: build a matrix from the vectors, reduce, inspect. Membership reads the augmented column, spanning reads the rows, redundancy reads the columns, and a basis needs both. Nothing here requires a technique the **reduction** does not already provide.